ABSTRACT

CyberSense AI: AI-Powered Human Cyber Behaviour Prediction System

Cybersecurity incidents are increasingly caused by unsafe human behaviour rather than failures in technical security mechanisms. Existing cybersecurity solutions primarily focus on detecting malicious websites, malware, or network attacks after they occur, while providing limited capability to predict the behavioural patterns that make individuals vulnerable to cyber threats. To address this challenge, we propose CyberSense AI, an AI-powered human cyber behaviour prediction system that employs Long Short-Term Memory (LSTM) networks to analyse sequential cybersecurity behavioural patterns and estimate an individual's future cyber risk. The proposed framework continuously monitors user cybersecurity activities over time, identifies behavioural drift, and predicts multiple categories of cyber risk, including phishing susceptibility, credential compromise, unsafe browsing behaviour, social engineering exposure, device security risk, and AI-enabled scam vulnerability. To improve transparency and user trust, the prediction process is integrated with Explainable Artificial Intelligence (XAI) techniques, enabling users to understand the key behavioural factors contributing to each risk prediction. The framework further incorporates adaptive thresholding, monitoring and category-wise behavioural summarization, allowing users with similar cybersecurity behaviour to receive personalized security insights and recommendations. By combining temporal behaviour modelling, explainable prediction, and proactive cyber risk assessment, the proposed system provides an intelligent and human-centred approach to cybersecurity awareness and prevention, with future work focusing on large-scale real-world validation and continuous model improvement.